Milestone-2 Report

ExoHabitAI: Data Preprocessing and Machine Learning Model Training

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Project: ExoHabitAI

Phase: Milestone-2 (Machine Learning Model Training)

Introduction

The ExoHabitAI project aims to predict the habitability potential of exoplanets using machine learning techniques. This milestone focuses on data preprocessing, feature preparation, model selection, training, evaluation, and final model selection.

Dataset Description

The dataset was obtained from the NASA Exoplanet Archive. Features include pl_eqt, pl_rade, and pl_orbsmax.

Data Preprocessing

Data cleaning, feature selection, habitability score generation, and target label creation were performed.

Problem Formulation

A supervised binary classification approach was used.

Dataset Preparation

The dataset was split into 80% training and 20% testing sets.

Model Selection

Logistic Regression was used as baseline and Random Forest as primary model.

Feature Scaling and Pipelines

StandardScaler and pipelines were used to avoid data leakage.

Model Training

Models were trained using default parameters and saved using joblib.

Model Evaluation

Accuracy, Precision, Recall, F1-score, and ROC-AUC were evaluated.

Hyperparameter Tuning

Default parameters were retained due to dataset size.

Model Comparison

Random Forest was selected as final model.

Habitability Ranking

Planets were ranked using predicted probabilities.

Model Interpretability

Feature importance was analyzed.

Results

The model achieved high accuracy due to clean dataset.

Limitations

Limited dataset size and features.

Conclusion

A complete ML pipeline was developed successfully.

Future Scope

Integration of larger datasets and advanced models.